



CIE A Level Chemistry



Your notes

5.2 Hess's Law

Contents

* Hess's Law



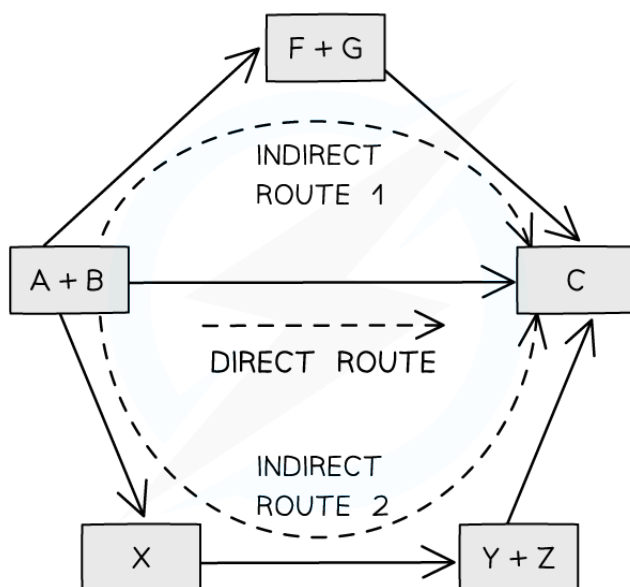
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Hess's Law

Hess Cycles

- Hess's Law states that:
"The total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place as long as the initial and final conditions are the same."
- This means that whether the reaction takes place in one or two steps, the total enthalpy change of the reaction will still be the same

Illustration of Hess's Law



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According to Hess' Law, the enthalpy change of the direct route, going from reactants ($A + B$) to product (C), is equal to the enthalpy change of the indirect routes

- Hess' Law is used to calculate enthalpy changes which can't be found experimentally using **calorimetry**, e.g.:

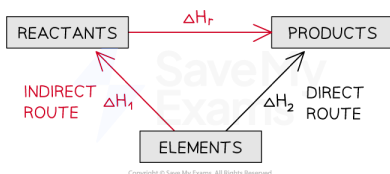


- ΔH_f (propane) can't be found experimentally as hydrogen and carbon don't react under standard conditions

Calculating ΔH_f from ΔH_c using Hess's Law energy cycles

- The products can be directly formed from the elements = ΔH_2
- OR
- The products can be indirectly formed from the elements = $\Delta H_1 + \Delta H_r$

Applying Hess's Law



The enthalpy change from elements to products (direct route) is equal to the enthalpy change of elements forming reactants and then products (indirect route)

Equation

$$\Delta H_2 = \Delta H_1 + \Delta H_r$$

Therefore,

$$\Delta H_r = \Delta H_2 - \Delta H_1$$



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Worked example

Calculate ΔH_f° for the following reaction:



The table shows the standard enthalpy of formations, ΔH_f° , relevant to this reaction.

Molecule	ΔH_f° kJ mol^{-1}
$\text{NaHCO}_3(\text{s})$	-950.8
$\text{Na}_2\text{CO}_3(\text{s})$	-1130.7
$\text{CO}_2(\text{g})$	-393.5
$\text{H}_2\text{O}(\text{l})$	-285.8

Answer

- Step 1:** Write the balanced equation at the top



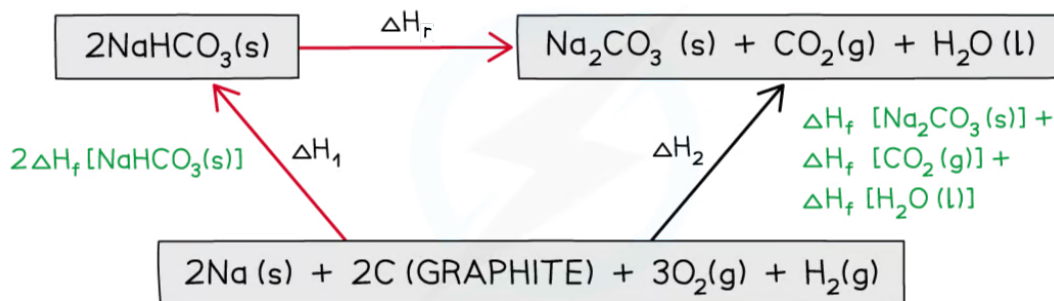
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- Step 2:** Draw the cycle with the elements at the bottom



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- Step 3:** Draw in all arrows, making sure they go in the correct directions. Write the standard enthalpy of formations



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▪ **Step 4:** Apply Hess's Law:

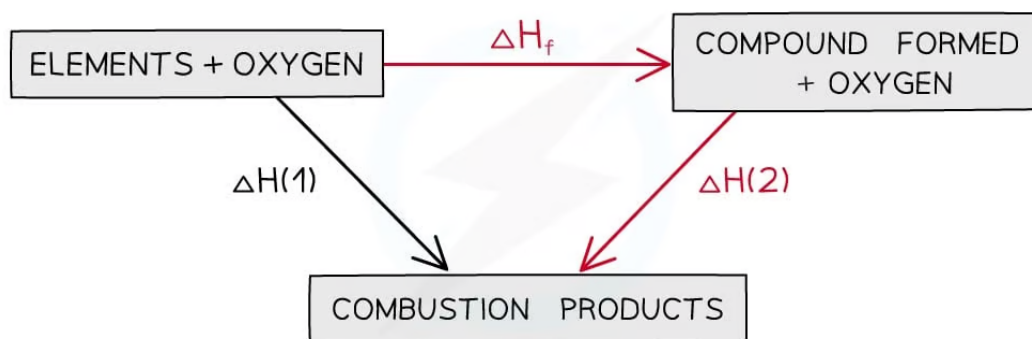
- $\Delta H_r = \Delta H_2 - \Delta H_1$
- $\Delta H_r = (\Delta H_f[\text{Na}_2\text{CO}_3(\text{s})] + \Delta H_f[\text{CO}_2(\text{g})] + \Delta H_f[\text{H}_2\text{O}(\text{l})]) - 2\Delta H_f[\text{NaHCO}_3(\text{s})]$
- $\Delta H_r = ((-1130.7) + (-393.5) + (-285.8)) - (2 \times -950.8)$
- $\Delta H_r = +91.6 \text{ kJ mol}^{-1}$

Calculating ΔH_f from ΔH_c using Hess's Law energy cycles

- The combustion products can be formed directly from elements to combustion products = ΔH_1
OR

The combustion products can be formed indirectly from elements to compounds to combustion products = $\Delta H_f + \Delta H_2$

Hess's Law and combustion enthalpies



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The enthalpy change going from elements to products (direct route) is equal to the enthalpy change of elements forming reactants and then products (indirect route)

- Equation

$$\Delta H_1 = \Delta H_f + \Delta H_2$$

Therefore,

$$\Delta H_f = \Delta H_1 - \Delta H_2$$



Your notes



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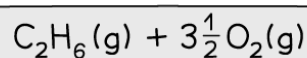
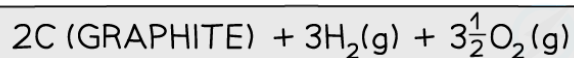
Worked example

Calculate, using the standard enthalpy change of combustion values given, the enthalpy of formation of ethane.

Reaction	ΔH_c kJ mol^{-1}
$\text{C (graphite)} + \text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g})$	-393.5
$\text{H}_2 (\text{g}) + \frac{1}{2}\text{O}_2 (\text{g}) \rightarrow \text{H}_2\text{O} (\text{l})$	-285.8
$\text{C}_2\text{H}_6 (\text{g}) + 3\frac{1}{2}\text{O}_2 (\text{g}) \rightarrow 2\text{CO}_2 (\text{g}) + 3\text{H}_2\text{O} (\text{l})$	-1559.7

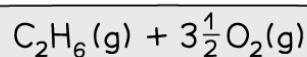
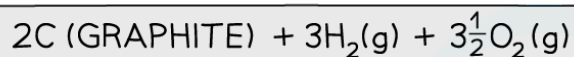
Answer

- Step 1:** Write the equation for the enthalpy change of formation at the top and add oxygen on both sides



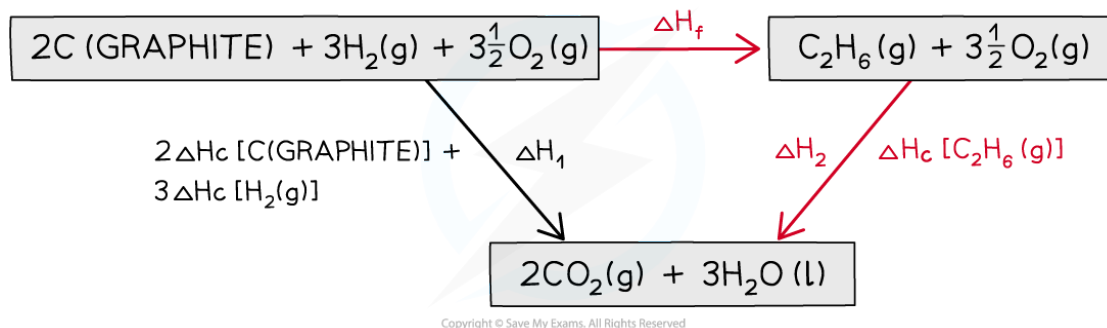
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- Step 2:** Draw the cycle with the combustion products at the bottom



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- Step 3:** Draw all arrows in the correct direction

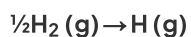


▪ **Step 4:** Apply Hess's Law:

- $\Delta H_f = \Delta H_1 - \Delta H_2$
- $\Delta H_f = ((2 \times -393.5) + (3 \times -285.8)) - (-1559.7)$
- $\Delta H_f = -84.7 \text{ kJ mol}^{-1}$

Calculating average bond energies using Hess's cycles

- Bond energies cannot be found directly so enthalpy cycles are used to find the **average bond energy**
- This can be done using enthalpy changes of **atomisation** and **combustion** or **formation**
- The **enthalpy change of atomisation** ($\Delta H_{\text{at}}^\theta$) is the enthalpy change when **one mole of gaseous** atoms is **formed** from its elements under standard conditions.
 - E.g. $\Delta H_{\text{at}}^\theta [\text{H}_2]$ relates to the equation:





Your notes

Worked example

Calculate the average bond enthalpy of the C-H bond, using the relevant ΔH_f^θ and ΔH_{at}^θ values shown in the table.

	Energy kJ mol^{-1}
$\Delta H_f^\theta [\text{CH}_4(\text{g})]$	-74.8
$\Delta H_{at}^\theta [\frac{1}{2}\text{H}_2(\text{g})]$	+218
$\Delta H_{at}^\theta [\text{C}(\text{graphite})]$	+717.7

Answer

- Step 1:** Write down the equation for the dissociation of methane at the top

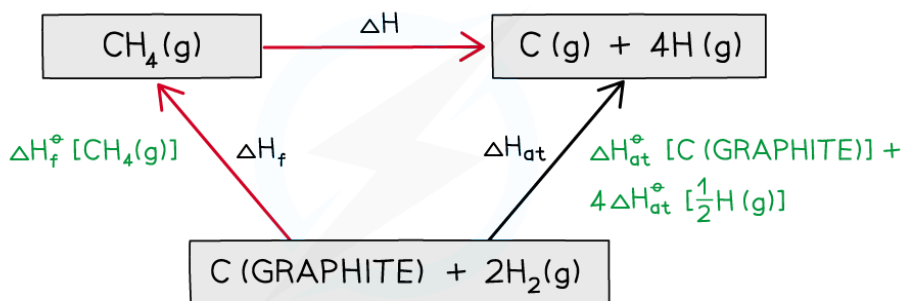


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- Step 2:** Write down the elements at the bottom



- Step 3:** Draw all arrows in the correct direction



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▪ **Step 4:** Apply Hess's Law:

- $\Delta H = \Delta H_{\text{at}}^\circ - \Delta H_f^\circ$
- $\Delta H = ((+717.7) + (4 \times 218)) - (-74.8)$
- $\Delta H_f^\circ = +1664.5 \text{ kJ mol}^{-1}$

▪ **Step 5:** Calculate the bond energy for **one** C-H bond

- There are 4 C-H bonds in methane
- Average bond enthalpy (C-H) = $\frac{+1664.5}{4}$
- Average bond enthalpy (C-H) = $+416 \text{ kJ mol}^{-1}$ (to 3 significant figures)

 Exam Tip

- Remember to take into account the number of moles of each reactant and product.
- For example, there are two moles of $\text{NaHCO}_3(\text{s})$ so the ΔH_f° value is multiplied by 2.